

Multi-Step Linear Equations with Rational Coefficients

Answer Key

Algebra 1 Equations and Inequalities

Quick Answers

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|---|----------------------------------|
| 1. 12 | 7. 7 |
| 2. 25 | 8. 4 |
| 3. 24 | 9. $-\infty$, -6, <i>closed</i> |
| 4. 16 | 10. $3x - 15$ |
| 5. 6 | 11. 5 |
| 6. 14, ∞ , <i>built-in function open</i> | 12. 13 |
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Curriculum

Level: Algebra 1 Equations and Inequalities

HSA-REI.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 12 tagged checks.

1. $\frac{x}{3} + 5 = 9$

Step 1. Undo the addition: subtract 5 from both sides, leaving $\frac{x}{3} = 4$.

Step 2. Undo the division: multiply both sides by 3, so $x = 12$.

Check. $\frac{12}{3} + 5 = 4 + 5 = 9 \checkmark$ $x = 12$

2. $\frac{2}{5}x - 3 = 7$

Step 1. Add 3 to both sides: $\frac{2}{5}x = 10$.

Step 2. Multiply both sides by the reciprocal $\frac{5}{2}$: $x = 10 \cdot \frac{5}{2} = 25$.

Check. $\frac{2}{5}(25) - 3 = 10 - 3 = 7 \checkmark$ $x = 25$

3. $\frac{x}{4} + \frac{x}{6} = 10$

Step 1. The denominators are 4 and 6, so the LCD is 12. Multiply *every* term by 12:

$$12 \cdot \frac{x}{4} + 12 \cdot \frac{x}{6} = 12 \cdot 10 \implies 3x + 2x = 120.$$

Step 2. Combine like terms: $5x = 120$.

Step 3. Divide both sides by 5: $x = 24$.

Check. $\frac{24}{4} + \frac{24}{6} = 6 + 4 = 10 \checkmark$ $x = 24$

4. $\frac{3}{4}(x - 8) = 6$

Step 1. The fraction multiplies the whole binomial, so undo it in one move: multiply both sides by $\frac{4}{3}$, giving $x - 8 = 8$.

Step 2. Add 8 to both sides: $x = 16$.

Check. $\frac{3}{4}(16 - 8) = \frac{3}{4}(8) = 6 \checkmark$ $x = 16$

5. $\frac{2}{3}x + 1 = \frac{x}{6} + 4$

Step 1. Denominators 3 and 6 give LCD 6. Multiply every term on *both* sides by 6:

$$4x + 6 = x + 24.$$

Step 2. Collect the variable on one side: subtract x from both sides, leaving $3x + 6 = 24$.

Step 3. Subtract 6, then divide by 3: $3x = 18$, so $x = 6$.

Check. Left: $\frac{2}{3}(6) + 1 = 5$. Right: $\frac{6}{6} + 4 = 5 \checkmark$ $x = 6$

6. $\frac{x}{2} - 3 > 4$

Step 1. Add 3 to both sides: $\frac{x}{2} > 7$.

Step 2. Multiply both sides by 2. Two is *positive*, so the symbol does not turn: $x > 14$.

Check. $x = 20$ gives $10 - 3 = 7 > 4 \checkmark$; $x = 14$ gives exactly 4, which is not greater, so 14 is correctly excluded. $x > 14$

7. $\frac{x + 2}{3} + \frac{x - 1}{2} = 6$

Step 1. LCD = 6. Multiplying a fraction such as $\frac{x + 2}{3}$ by 6 multiplies the *whole* numerator, so keep the parentheses:

$$2(x + 2) + 3(x - 1) = 36.$$

Step 2. Distribute: $2x + 4 + 3x - 3 = 36$.

Step 3. Combine like terms: $5x + 1 = 36$, so $5x = 35$ and $x = 7$.

Check. $\frac{9}{3} + \frac{6}{2} = 3 + 3 = 6 \checkmark$ $x = 7$

8. $\frac{3}{4}x - \frac{2}{3} = \frac{x}{6} + \frac{5}{3}$

Step 1. Denominators 4, 3 and 6 give LCD 12. Multiply every term by 12:

$$9x - 8 = 2x + 20.$$

Step 2. Subtract $2x$ from both sides: $7x - 8 = 20$.

Step 3. Add 8, then divide by 7: $7x = 28$, so $x = 4$.

Check. Left: $3 - \frac{2}{3} = \frac{7}{3}$. Right: $\frac{4}{6} + \frac{5}{3} = \frac{2}{3} + \frac{5}{3} = \frac{7}{3} \checkmark$ $x = 4$

9. $-\frac{2}{3}x + 1 \geq 5$

Step 1. Subtract 1 from both sides: $-\frac{2}{3}x \geq 4$.

Step 2. Multiply both sides by $-\frac{3}{2}$. The multiplier is *negative*, so the inequality symbol reverses: $x \leq -6$.

Check. $x = -9$ gives $6 + 1 = 7 \geq 5$ ✓, while $x = 0$ gives $1 \geq 5$, which is false — so the solution really is the interval below -6 . $x \leq -6$

10. Clear the fractions from $\frac{5}{6}x - \frac{1}{2} = \frac{1}{3}x + 2$.

Step 1. The denominators are 6, 2 and 3, so the LCD is 6.

Step 2. Multiply *each of the four terms* by 6:

$$6 \cdot \frac{5}{6}x - 6 \cdot \frac{1}{2} = 6 \cdot \frac{1}{3}x + 6 \cdot 2.$$

Step 3. Each product is an integer term, which gives the equivalent integer equation below. It is equivalent because multiplying both sides of an equation by the same nonzero number never changes its solution set — both equations are true exactly when $x = 5$.

$5x - 3 = 2x + 12$

11. $\frac{1}{2} \left(\frac{x}{3} - 4 \right) + \frac{5}{6}x = 3$

Step 1. Distribute the $\frac{1}{2}$ across the inner parentheses: $\frac{x}{6} - 2 + \frac{5}{6}x = 3$.

Step 2. The two x -terms already share the denominator 6: $\frac{x}{6} + \frac{5x}{6} = \frac{6x}{6} = x$, so the equation becomes $x - 2 = 3$.

Step 3. Add 2 to both sides: $x = 5$.

Check. $\frac{1}{2} \left(\frac{5}{3} - 4 \right) + \frac{5}{6}(5) = \frac{1}{2} \left(-\frac{7}{3} \right) + \frac{25}{6} = -\frac{7}{6} + \frac{25}{6} = 3$ ✓ $x = 5$

12. $\frac{3(2x-1)}{4} - \frac{2(x+1)}{3} = \frac{5}{12}x + 4$

Step 1. Denominators 4, 3 and 12 give LCD 12. Multiply every term by 12, keeping each numerator inside its parentheses:

$$9(2x - 1) - 8(x + 1) = 5x + 48.$$

Step 2. Distribute on the left: $18x - 9 - 8x - 8 = 5x + 48$.

Step 3. Combine like terms: $10x - 17 = 5x + 48$.

Step 4. Subtract $5x$ and add 17: $5x = 65$, so $x = 13$.

Check. Left: $\frac{3(25)}{4} - \frac{2(14)}{3} = \frac{75}{4} - \frac{28}{3} = \frac{225 - 112}{12} = \frac{113}{12}$. Right: $\frac{65}{12} + 4 = \frac{113}{12}$ ✓

$x = 13$